

Vortex transport and screening in a closed two-dimensional Bose field

Friction, transverse force, diffusion and the finite-wavevector dielectric relation of quantized vortices, measured in an energy-conserving classical field with no reservoir and no fit parameter — pre-registered, with the instrument failures that preceded the measurements

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SocrateAI-Scientific-QuantumFluids

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Abstract

A quantized vortex in a thermal two-dimensional Bose gas is dragged by the normal fluid (friction α), pushed sideways by it (transverse coefficient α') and kicked by it (diffusion η). The only first-principles values of α for a bosonic classical field stop at $0.18 T_{\text{BKT}}$ and carry no error bar on α' [1]; the fluctuation–dissipation relation tying η to α was proposed for test [2] and found violated by two orders of magnitude in one experiment [3]. We measure the three coefficients in the projected Gross–Pitaevskii equation — microcanonical, no damping term, no noise put in by hand — with sub-grid vortex tracking and estimators that are identities of the dissipative point-vortex model, machine-checked in Lean 4. Every hypothesis, gate and decision rule was committed before the corresponding run; the instrument failed its $T = 0$ control twice before it passed, an earlier gate verdict built on the flawed instrument is withdrawn, and both are reported. Results: at $T/T_{\text{BKT}} = 0.14, 0.27, 0.43$, $\alpha = 0.0062(4), 0.014(2), 0.02(1)$ — size- and estimator-independent, $\alpha \approx 0.24 \rho_n/\rho$; α' is zero within 1% once the $T = 0$ compressible pair speed is subtracted, excluding the Iordanskii value ρ_n/ρ at 18σ (the registered Iordanskii-like prediction is refuted); the Einstein relation is inconclusive by the registered rule, with the one admitted temperature at $\eta K/\alpha = 2.1 \pm 0.7$ and the others at 1.4–3 — not 100. Single pairs stop shrinking while the phonon band takes the impulse they shed (a self-generated counterflow, with a published precedent [4]); the decisive box-size control is not run. Separately: the single non-equilibrium pair that corrupts the stiffness estimator in finite boxes decays by $4\text{--}27\times$ given L^z -scaled time, one of six $L = 192$ trajectories still unfinished at $t = 14\,500$; the pre-registered mode-by-mode dielectric relation *fails* in the thermal pair gas (coherence 0.06–0.28) while the vortex-charge structure factor equals the Kosterlitz–Thouless polarisability Σd^2 from the measured pair sizes to 5–21%; and the Josephson offset $\eta_{g_1} K - 1 = +0.3$ survives at $L = 192$ in equilibrium, independent of window and torus geometry. No physics is claimed as new; the numbers, the refutations and the method are.

1 What is measured and why it is open

With the normal fluid at rest a vortex of charge q moves at

$$\mathbf{v} = (1 - \alpha') \mathbf{v}_s - \alpha q \hat{z} \times \mathbf{v}_s, \quad (1)$$

$\mathbf{v}_s$ the superfluid velocity induced at its position by the other vortices [5, 6]. For $\alpha \rightarrow 0$ the two sides of a long controversy predict different α' : no transverse force from the normal fluid, $\alpha' = 0$ [7], or the Iordanskii force, $\alpha' = \rho_n/\rho$ [5, 6]. If the field is the vortex’s heat bath, friction

and diffusion obey $\eta = \alpha T / (2\pi\rho)$ ($\hbar = m = k_B = 1$) [2]. For a bosonic classical field, $\alpha(T)$ exists only below $0.18 T_{\text{BKT}}$ [1], α' changes sign within its noise there and “deviates” in 3D Kelvin-wave fits [8]; the Einstein relation has never been tested in a closed field; helium-film analyses assume it [9]; the one cold-atom measurement of η [3] is $100\times$ the prediction. A review of the field is [10].

A second question is static. The superfluid stiffness is read from the transverse current response, $\rho_n(k) = \langle |J_T(k)|^2 \rangle / (TL^2)$; in the Coulomb-gas picture vortices reduce it through their polarisation [11, 12, 13]. For a compressible Bose field no functional form at finite k exists: classical-field work extrapolates to $k \rightarrow 0$ by fits of convenience and has reported negative superfluid fractions attributed to noise [14, 15].

Model and assets. Projected Gross–Pitaevskii equation ($\hbar = m = g = 1$, $dx = \xi/2$, cutoff at half the grid wavenumber), energy conserved to 10^{-7} – 10^{-6} over every run; boxes $L = 64$ (transport) and $L = 192$ (equilibration, dielectric relation); a calibrated two-window thermometer; $T_{\text{BKT}}(L=64) = 0.821$ from the round-2 ladder; thermal states with $T/T_{\text{BKT}} = 0.14, 0.27, 0.43$ and no thermal vortex (raw count ≤ 0.36), and one at 0.56 with thermal pairs (report only). Pre-registrations, amendments and gate records are files in the repository (docs/designs/PGPE_FRICTION_PREREG.md and amendments A1, A1.1, A1.2; PGPE_ALPHAPRIME_PREREG.md; PGPE_EINSTEIN_PREREG.md; PGPE_DIELECTRIC_PREREG.md with amendments D-A1, D-A2), each committed before the data it governs existed.

2 The instrument, and what it took to trust it

Estimators (identities of the model). Writing the configuration’s point-vortex Hamiltonian H (Weiss–McWilliams on the torus [16]), Eq. (1) is a skew part plus $-\frac{\alpha}{2}\nabla H$, so $\dot{H} = -\frac{\alpha}{2}|\nabla H|^2$ for any configuration and any α' : $\alpha = -\Delta H / (2\int \sum_i |\mathbf{v}_{s,i}|^2 dt)$ is the *energy estimator*, pairing-independent and blind to α' (*energy_dissipation*). The displacement regressed on the two orthogonal predictors $\int \mathbf{v}_s dt$ and $-q \int \hat{z} \times \mathbf{v}_s dt$ gives $1-\alpha'$ and α separately (*centre_velocity_identity*); its residuals’ mean square against lag gives η , quoted only if the scaling exponent lies in $[0.8, 1.2]$. For a stochastic pair with drift $-2\alpha\mathbf{d}/|\mathbf{d}|^2$ and diffusion $D = 2\eta$, the Kosterlitz–Thouless pair distribution $|\mathbf{d}|^{-K}$ carries zero probability current iff $DK = 2\alpha$ (*zero_flux_iff*); the Einstein statistic is $R_E = \eta K / \alpha$.

Gates. Synthetic Langevin trajectories with known (α, α', η) are recovered to 7%, 0.002 and 25% (G0; its first attempt failed on a generator defect, on file). The $T = 0$ control (G1) failed on its first run and exposed two defects of the imprint used by this programme since round 2: the theta-function phase is periodic only when $\sum \mathbf{q}\mathbf{r} \in L\mathbb{Z}^2$ — a single pair carries a phase step $2\pi d/L$ along a whole line — and the amplitude factor $[r^2/(r^2 + 2)]^{1/2}$ per vortex has a $1/r^2$ tail four times the Bernoulli depletion, whose sound drifts pair separations by 0.6–0.8 in 400 time units at $T = 0$: an apparent $|\alpha|$ of 0.007–0.010, as large as the thermal value at $0.14 T_{\text{BKT}}$ (cf. [17] on product-ansatz dipoles). With a periodic phase and a Bernoulli amplitude (*imprint_v2*) G1 passes: $|\hat{\alpha}| = 4 \times 10^{-4}$, $1 - \hat{\alpha}' = 0.9986$, no diffusion, single-pair field momentum constant to 3×10^{-10} and 0.96–0.99 of $2\pi nd$, separation constant at 9.82 — a pair at $T = 0$ neither shrinks nor diffuses [18]. The thermal known answer (G2, $T/T_{\text{BKT}} = 0.14$, two antiparallel pairs so that no net impulse is shed) gives $\alpha = 0.00705 \pm 0.00012$ inside the window set from [1] before the run.

Withdrawn. An earlier known-answer gate of this study (single dipoles, the old imprint, a coarse-grained detector biased low by ≈ 1 in separation) was reported as “PASS, narrowly” with an unexpected dependence of α on pair size and a stalled pair. All three were artefacts of that instrument and are withdrawn (LEDGER CLAIM-078). The stall motivated a hypothesis — the impulse shed by a shrinking pair in a closed box drives a phonon wind that stalls it (`wind_stall`) — which stays registered; its status is in Section 4.

3 Friction, transverse coefficient, diffusion

The coldest temperature, in detail. Eight runs at $T/T_{\text{BKT}} = 0.14$ (antiparallel pairs, $d_0 = 8$ and 12, 2000 time units, corrected imprint, raw detection with sub-grid refinement):

	$d_0 = 8$	$d_0 = 12$	all
α , energy estimator	0.0057 ± 0.0006	0.0063 ± 0.0011	0.0062 ± 0.0004
α , regression	0.0059 ± 0.0005	0.0063 ± 0.0010	0.0065 ± 0.0004
α' , raw (before the $T = 0$ baseline of the next paragraph)	-0.014 ± 0.002	-0.004 ± 0.002	-0.010 ± 0.002

Friction is $17\times$ the $T = 0$ floor, within a factor two of [1] interpolated to this temperature and of an independent configuration of this programme (eight imprinted vortices, $\alpha = 0.0056$). The raw α' differs between the two pair sizes by five standard errors — the first sign that it contains a size-dependent $T = 0$ term, treated below.

Three temperatures. The first production pass at $T/T_{\text{BKT}} = 0.27, 0.43, 0.56$ produced no data (thermal jitter of the detected core exceeded the tracking radius inside the settling window; amendment A1.2, radius 3.0, nothing else changed) and was repeated. Friction, energy estimator (regression in parentheses):

T/T_{BKT}	ρ_n/ρ	α	$\alpha(d_0=8)/\alpha(d_0=12)$	$\alpha/$
0.14	0.027	0.0062 ± 0.0004 (0.0065)	0.91	
0.27	0.053	0.0138 ± 0.0023 (0.0156)	0.76	
0.43 (base flagged: thermal vortices in $> 5\%$ of samples)	0.094	0.021 ± 0.014 (0.024)	1.04	

The three registered criteria pass: no intrinsic size dependence (ratios in $[0.7, 1.43]$), monotone in T , the two estimators within 25%. Not pre-registered, and the cleanest statement of the data: $\alpha \approx 0.24 \rho_n/\rho$ at all three temperatures. Against the literature, 0.0062 at $0.14 T_{\text{BKT}}$ is a factor two below the interpolation of [1] (whose T_{BKT} is itself uncertain); at 0.27 and 0.43 no closed-field value existed, and ours lie in the range of the atomic-gas measurements [19] and above the unitary-gas value [20].

Transverse coefficient. Registered estimator: weighted regression of $\hat{\alpha}'$ on ρ_n/ρ over the three temperatures and $T = 0$. Values $+0.0014$, -0.0100 ± 0.0015 , -0.0154 ± 0.0029 , -0.0093 ± 0.0106 ; slope $b = -0.31 \pm 0.05$. By the rule, “no transverse force” (the Iordanskii window $b \in [0.6, 1.4]$ is excluded at 18σ); **the registered prediction — Iordanskii-like — is refuted.** The raw α' is negative at 7σ and 5σ , which neither theory predicts; two post-hoc controls (estimator unchanged in the verdict) account for it. In a periodic box a single pair carries net momentum, so the fluid has a mean velocity $u = P/nL^2$ absent from the point-vortex prediction; in the fluid frame a $T = 0$ pair moves at the torus point-vortex speed to $+0.4\%$ for $d \geq 8$, $+1\%$ at $d = 6$ and $+3\text{--}4\%$ at $d = 3\text{--}4$ (the Jones–Roberts compressible correction; production runs have $|u| \leq 2 \times 10^{-3}$ and two counter-moving pairs, so the frame correction to them is ≤ 0.0002). Subtracting that $T = 0$ baseline at the matched pair sizes leaves $\alpha' = -0.006$, -0.011 , -0.006 with a mapping

uncertainty of the same order: $\alpha' = 0$ **within about** 1%, and the Iordanskii values +2.7%, +5.3%, +9.4% are excluded. The round-3 configuration (eight imprinted vortices, $d = 16$) gives $1 - \alpha' = 1.0012$, consistent.

Diffusion and the Einstein relation. η from the residuals of the two-coefficient regression against lag, quoted only where the scaling exponent lies in $[0.8, 1.2]$ (rule E2): 3.7×10^{-4} (exponent 0.71, not quoted), $(1.08 \pm 0.35) \times 10^{-3}$ (0.82, quoted), $(1.8 \pm 0.4) \times 10^{-3}$ (1.28, not quoted). With $K = 2\pi n_s/T$, $R_E = \eta K/\alpha = 3.2$, **2.1 ± 0.7** , 1.4. **Verdict by the registered rule: inconclusive** (one admitted temperature; two required). Informally and against the rule, the vortex diffuses at 1.4–3 times the Einstein value, decreasing with temperature — not the 100× of [3] and not clearly 1. The exponents are the reason for caution, and were examined after the fact (amendment E-A1, filed after the analysis it concerns and labelled a retrodiction): over lags 20–400 the exponent rises with T ($0.71 \rightarrow 0.82 \rightarrow 1.28$), but over lags ≥ 100 the two cooler temperatures give 1.1–1.2; the sub-diffusion at $0.14 T_{\text{BKT}}$ is a short-lag regime, not a bounded orbit. The machine-checked statement behind that reading is one-directional: a finite quasi-periodic sum has a bounded mean-square increment, so growth refutes a regular island and a plateau never proves one (`QuasiPeriodicBound.lean`).

4 The phonon wind of a closed box

A pair of separation d carries impulse $2\pi\rho_s d$; in a periodic box its decrease has nowhere to go but the phonons, which then drift at $u = 2\pi\rho_s(d_0 - d)/(\rho_n L^2)$ and reduce the friction drive, $\dot{d} = -2\alpha(1/d - u)$, stalling at $b = [d_0 + (d_0^2 - 2\rho_n L^2/\pi\rho_s)^{1/2}]/2$ when the root is real (`wind_stall`). At $T/T_{\text{BKT}} = 0.14$, $L = 64$, $d_0 = 12$ should stall ($b = 10.2$ in the plane formula, ≈ 9 torus-corrected) and $d_0 = 8$ should not. Two single-pair runs of 4000 time units: neither pair annihilates; one shrinks $11.6 \rightarrow 9.3$ and stays there (late d^2 slope 4% of the early one: criterion W1 met), the other shrinks $11.2 \rightarrow 10.2$ and re-expands to 11.2 (W1 failed by the letter, the pair moving back toward the stall point); the momentum of the modes with $|k| > 1$ follows the impulse the pair sheds with slopes 0.68 and 0.52 (W3, window $[0.5, 1.2]$, met), falling back when the pair re-expands. Zero-impulse pairs at the same temperature shrink steadily from 10 to 7–8. The decisive control — the same pair in an $L = 96$ box, where no stall is predicted — has not been run; the hypothesis is **unresolved with first support**. Its object has a precedent: in the truncated GPE a thermal cloud with net momentum relative to the condensate is a metastable state decaying by Arrhenius ring nucleation [4]; ours would be the same counterflow, self-generated by a shrinking pair. On the torus the zero-circulation dipole is a relative equilibrium of the point-vortex flow [21], so the wobble of the stalled pair is bath-driven by theorem; its spectrum over $t \in [2000, 4000]$ puts only 32–36% of the variance in three lines, led by the slowest resolvable frequency: a stalled equilibrium perturbed by a bath, not a quasi-periodic orbit. If the mechanism holds, closed-box friction measurements on single pairs at low temperature are contaminated unless $\rho_n L^2 \gg \pi\rho_s d_0^2/2$ — a condition the literature’s truncated-GPE measurements should be checked against.

5 Box-scale vortex charge, equilibration and the stiffness estimator

Seven of eighteen large-box runs of an earlier round reached $n_s/n < 0$ with the same vortex number as healthy partners. A pre-registered test on their vortex positions refuted the Onsager-cluster reading (no like-sign clustering, positive vortex temperature, 98–99.7% of vortices paired with an opposite neighbour); post hoc, the point-vortex transverse current $n_{\text{eff}}^2 (2\pi)^2 |\rho_q(k)|^2 / k^2$ computed from positions alone tracks the measured one across all eighteen runs with $r = 0.996$: the

anomaly is one $O(1)$ unscreened pair at the box scale — the finite- k form of the polarisation term of [11, 12], the single free vortex of early superconducting films, and the post-quench surviving pair that Chu and Williams asked simulations to look for [22]. Such a pair has Boltzmann weight $(L/\xi)^{-K} \lesssim 10^{-9}$: it is a remnant of the initial relaxation. Equilibrium finite-size stiffness is monotone in L [23, 24]; coarsening under conservative dynamics gives $z \approx 1.7$ – 1.8 [25] ($L^2 \ln L$ for dissipative XY [26, 27]; $L^2/\ln L$ Hamiltonian XY [28]), so the warm-up that cured $L = 128$ should scale to $\approx 11\,000$ – $14\,600$ at $L = 192$. Registered before the runs: continuing the six $L = 192$ trajectories to $t = 14\,500$ should admit at least five of six (E1), bring the vortex transverse current below 0.45 in every admitted run (E2), and drop it by at least $2\times$ in the two anomalous ones (E3):

run	admitted	R_T^{vortex} : [6000, 7000] $\rightarrow$ [13 500, 14 500]	n_s/n
e1.00 s11 (anomalous)	yes	$1.98 \rightarrow 0.07$ ($\div 27$)	$-0.38 \rightarrow +0.81$
e1.00 s12	yes	$0.65 \rightarrow 0.08$	$+0.49 \rightarrow +0.77$
e1.10 s11 (anomalous)	yes	$3.38 \rightarrow 0.80$ ($\div 4.2$)	$-0.96 \rightarrow +0.34$
e1.10 s12	yes	$0.74 \rightarrow 0.19$	$+0.41 \rightarrow +0.80$
e1.20 s11	yes	$0.55 \rightarrow 0.38$	$+0.56 \rightarrow +0.61$
e1.20 s12	yes	$0.44 \rightarrow 0.41$	$+0.73 \rightarrow +0.74$

E1 pass (6/6), E3 pass, E2 fail on one run (e1.10 s11, still at 0.80 with $K = 3.5$ against its partner’s 8.3): the box-scale charge decays as the mechanism says in every run, and the equilibration time of an $L = 192$ box at this temperature exceeds 13 500 for some trajectories — above the L^2 and $z \approx 1.7$ estimates, as the $L^2 \ln L$ form would have it; one trajectory is one trajectory. The offsets $\eta_{g_1} K - 1$ on the equilibrated window are $+0.20, +0.16, +0.30$ ($e = 1.00, 1.10, 1.20$): positive at every size and energy once equilibrated, so the earlier “saturation” verdict rested on under-equilibrated states, and the $L = 128$ states fall under the same suspicion.

Josephson relation. On these equilibrated $L = 192$ states the coherence exponent fitted with the exact torus Green function on $r \in [8, 32]$ gives $\eta_{g_1} K - 1 = +0.31 \pm 0.11$, unchanged by the fit window (16–48) and equal to the naive-window value: the 12–27% excess reported at $L \leq 64$ is neither a geometry factor (cf. “close to unity” [29]) nor a finite-size transient. It persists in equilibrium at $L = 192$. What remains is the estimator pair itself — η from g_1 against K from current correlators on a fixed-momentum microcanonical field, for which no fluctuation formula has been derived — or the renormalisation of the bare stiffness by amplitude fluctuations. Reported, not explained; the earlier statement that this excess is “unexplained finite-size” is amended.

6 The finite-wavevector dielectric relation

Pre-registered claims: mode by mode and snapshot by snapshot, $J_T(k) = n_{\text{eff}} (2\pi i \rho_q(k)/|k|)$ plus a phonon part uncorrelated with ρ_q (D1: coherence $\gamma^2 \geq 0.4$ on every shell $|m|^2 \leq 16$ in ≥ 5 of 6 runs), with one real k -independent n_{eff} (D2); and (D4) the plateau of $\langle |\rho_q(k)|^2 \rangle / k^2$ equal to $\Sigma_{\text{pairs}} d^2/2$ from the minimum-cost matching of the detected vortices, within 30%. Five gate runs on imprinted pair gases failed on the amplitude’s known answer and are on file; the coherence criterion passed in all five ($\gamma^2 = 1.000$ at $T = 0$ with the corrected imprint): the relation exists for an imprinted pair gas, with an amplitude that rises with pair size (0.922, 0.971, 0.991 at $d = 5, 8, 14$ — a tight pair carries less current than two point vortices). On the six equilibrated $L = 192$ states (170–830 vortices, almost all in tight pairs, $T/T_{\text{BKT}} \approx 0.6$ – 0.85): **D1 fails (1 of 6), D2 fails (1 of 6), D4 passes (0.79–0.95 in all five runs without a box-scale pair).** The coherence between the transverse current and the detected charge density at the same k and t is 0.06–0.28; the one run that passes is the one still carrying a box-scale pair. So the $r = 0.996$ of the diagnosis above and the 0.987 of the triage probe were, as the pre-registration

itself warned, correlations of run-averaged powers carried by box-scale pairs: the diagnostic detects one unscreened pair and does not extend to a dielectric function of the thermal pair gas. What survives is D4: the bound-pair gas has a k -independent charge structure factor out to $k\xi \approx 0.13$ whose value is the Kosterlitz–Thouless polarisability $\varepsilon - 1 \propto \Sigma d^2$ computed from the measured pair sizes, 5–21% lower (screening of dipoles by dipoles), read directly from vortex positions in a Bose field [30, 31, 32]. How much of the stiffness that polarisability accounts for is what D1’s failure leaves open; the first analysis code carried a k^2 unit error in D4, corrected to the registered formula before anything else was changed.

7 What this establishes, and what it does not

Established (pre-registered criteria, or controls stated as such).

- At $T = 0$ a vortex pair neither shrinks nor diffuses over 400 time units and moves at the torus point-vortex velocity to 0.14% in the fluid frame; tight pairs ($d \leq 6$) move 1–4% faster (compressible correction).
- $\alpha(T) = 0.0062(4), 0.014(2), 0.02(1)$ at $T/T_{\text{BKT}} = 0.14, 0.27, 0.43$, size- and estimator-independent; $\alpha \approx 0.24 \rho_n / \rho$.
- α' is not Iordanskii-like (18σ); after the $T = 0$ baseline it is zero within $\sim 1\%$.
- The depressed-stiffness states of finite boxes carry one unscreened box-scale pair, not clusters; given L^z -scaled time it decays by $4\text{--}27\times$ (six of six admitted; one trajectory not finished at 14 500).
- The vortex-charge structure factor of the thermal pair gas is k -independent and equals the Kosterlitz–Thouless polarisability from the measured pair sizes to 5–21%.
- The Josephson offset $\eta_{g1} K - 1 = +0.31 \pm 0.11$ survives at $L = 192$ in equilibrium, independent of window and torus geometry.
- Theorems (standard axioms): energy dissipation of any skew-plus-gradient vortex flow, the d^2 law and lifetime bound, the centre-velocity identity, the wind stall, the zero-flux/Einstein equivalence, the pair-polarisation form and response ceiling, the charge-density/matching bound, the Kosterlitz-flow invariant, and the bounded increments of quasi-periodic signals.

Refuted, inconclusive, or not established.

- *Refuted*: the registered Iordanskii-like prediction for α' ; the mode-by-mode dielectric relation in the thermal pair gas; the Onsager-cluster reading of the depressed-stiffness states; the earlier “saturation” reading of the finite-size offset.
- *Inconclusive by rule*: the Einstein relation (one admitted temperature; $1.4\text{--}3\times$ informally).
- *Unresolved*: the phonon-wind mechanism (first support; $L = 96$ control missing); the cause of the Josephson offset; the current carried by tight thermal pairs; why $\alpha \propto \rho_n$ with coefficient 0.24.
- *Scope*: a Rayleigh–Jeans field with one cutoff, $mg = 1$, two dimensions; nothing about quantum gases or helium follows without a cutoff study.

Method, in one paragraph. The point-vortex model is a variational projection of the field onto the manifold of vortex positions; the coefficients measured here are the statistics of the

field's leak out of that manifold — the same object that a recent many-body experiment reads from the other end, by projecting a quantum state back onto a variational manifold at every step and asking how much coherence survives [33]. The hypotheses tested here were themselves ranked by an automated five-minute-probe loop under priors fixed before any probe ran, and the literature behind the design is a local database of 774 entries (382 arXiv identifiers verified against the arXiv API), described in the repository.

Process record. Three gate failures on the transport instrument (one generator, two imprint), one withdrawn gate verdict, one tracking amendment after a production pass with no data; five dielectric gate failures on the amplitude's known answer with the coherence criterion untouched; one registered prediction refuted, one inconclusive by rule, one amendment filed after the analysis it concerned and labelled a retrodiction; one unit error in analysis code caught and corrected before use. Data, code, pre-registrations, the ledger of claims and the Lean sources are in the repository.

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